

The bacterial epigenome

María A. Sánchez-Romero and Josep Casadesús  *

Abstract | In all domains of life, genomes contain epigenetic information superimposed over the nucleotide sequence. Epigenetic signals control DNA–protein interactions and can cause phenotypic change in the absence of mutation. A nearly universal mechanism of epigenetic signalling is DNA methylation. In bacteria, DNA methylation has roles in genome defence, chromosome replication and segregation, nucleoid organization, cell cycle control, DNA repair and regulation of transcription. In many bacterial species, DNA methylation controls reversible switching (phase variation) of gene expression, a phenomenon that generates phenotypic cell variants. The formation of epigenetic lineages enables the adaptation of bacterial populations to harsh or changing environments and modulates the interaction of pathogens with their eukaryotic hosts.

Methylome

Overall DNA methylation pattern in a genome.

Adaptive value

In population genetics, the contribution of a phenotypic trait to the fitness of an individual or a population.

Restriction–modification (R–M) systems

Machine systems that eliminate foreign DNA molecules by endonucleolytic cleavage and protect the genome by modification of the cognate endonuclease target. A frequent type of protection is DNA methylation.

In the second half of the 20th century, eukaryotic genomes were found to contain information superimposed over the nucleotide sequence¹. In multicellular eukaryotes, this additional epigenetic information controls differentiation and development, and its perturbation can cause pathological conditions². The main epigenetic signals in eukaryotic genomes are DNA methylation³ and post-translational modification of histones⁴, and both mechanisms contribute to establish the transcription patterns that govern the formation of specialized cell types. DNA methylation controls the binding of proteins to DNA and provides a primary layer of information to shape the eukaryotic epigenome³. DNA methylation is also found in bacterial genomes, where it controls DNA–protein interactions as in eukaryotes^{5–8} (BOX 1).

The methylated bases found in bacteria are N⁶-methyladenine (6mA) and C⁵-methyl-cytosine (5mC), which also exist in eukaryotes, and N⁴-methyl-cytosine (4mC), which is found only in bacteria and archaea (TABLE 1). Base methylation is post-replicative, and the bacterial genome is methylated at both DNA strands in the absence of DNA replication. Upon DNA replication, base methylation is present in one strand only (hemimethylation)⁵. Early studies in *Escherichia coli* and other gammaproteobacteria unveiled roles for DNA adenine methylation in chromosome replication, mismatch repair, control of transposon activity and regulation of transcription^{5,8}. Proteins that can discriminate the methylation state (methylated or hemimethylated) of cognate binding sites were also identified⁵. In alphaproteobacteria, DNA adenine methylation was found to govern the cell cycle⁹. Seminal studies in uropathogenic *E. coli* unveiled a role for DNA adenine methylation in the formation of phenotypic variants of bacterial cells¹⁰, in a manner reminiscent of the involvement of DNA cytosine methylation in the formation of eukaryotic

cell lineages. At the turn of the century, DNA adenine methylation was found to be essential for the pathogenesis of *Salmonella enterica* subsp. *enterica* serovar Typhimurium^{11,12}, and later of other pathogens^{13,14}. Together, these studies extended to bacteria the notion that DNA methylation can exert epigenetic control over the phenotype^{6,7,15–17}. The possibility that phosphorothioation of the DNA backbone may provide additional epigenetic marks, perhaps in a manner complementary to DNA methylation, remains open^{18,19}.

A hurdle in the investigation of the roles of DNA methylation in bacteria was that most studies focused on 6mA, which is a base modification that was difficult to detect. This technical limitation has been overcome through the development of DNA-sequencing technologies that enable the high-throughput analysis of DNA methylation across the genome (the methylome). Two such technologies, single-molecule real-time sequencing (SMRT)^{20,21} and nanopore sequencing²², are currently ‘gold standards’ in the study of bacterial methylomes²³. SMRT technology permits researchers to infer the presence of methylated bases on the template DNA strand from kinetic signatures that are specific for each type of methylated base²⁰. Nanopore sequencing identifies methylated nucleotides by detecting changes in electric current density upon electrophoretic passage of the sample through an orifice²². Methylome analysis has shown that base methylation is widespread in the genomes of bacteria and archaea²⁴, including the small genomes of certain (but not all) obligate parasites²⁵. The wealth of information provided by methylome analysis has also updated (and sometimes changed) classic notions about the physiological relevance and adaptive value of DNA methylation in bacteria. For instance, the DNA methyltransferases of restriction–modification (R–M) systems were thought to function in genome defence

Departamento de Genética,
Facultad de Biología,
Universidad de Sevilla,
Sevilla, Spain.

*e-mail: casadesus@us.es

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Box 1 | The epigenomes of eukaryotes and bacteria

- In eukaryotes, epigenetic modification of the genome involves DNA methylation³ and histone modification⁴. Bacteria lack histones, and epigenetic control relies on DNA methylation only⁶.
- In eukaryotes, de novo and maintenance forms of DNA methylation are performed by separate enzymes². Bacterial DNA methyltransferases have both de novo and maintenance activities³⁷.
- In eukaryotes, two main mechanisms exist to erase DNA methylation marks: active demethylation by dedicated proteins (Tet enzymes), and passive demethylation by the hindrance of DNA methylase activity upon DNA replication³⁵. In bacteria, DNA demethylation is usually passive⁶⁶, and the relevance of active demethylation by DNA repair remains to be evaluated⁴⁸².
- In both bacteria and eukaryotes, transcriptional repression by DNA methylation is common^{3,6}. Transcriptional activation of bacterial genes under DNA methylation control often involves demethylation (partial or complete, single- or double-stranded) of promoters or regulatory regions^{57,72,89,90,94,158}.
- The methylated base typically involved in the control of eukaryotic transcription is C⁵-methyl-cytosine³, whereas in bacteria it is often N⁶-methyl-adenine^{7,14}. However, direct control of bacterial transcription by C⁵-methyl-cytosine has been demonstrated recently¹²⁶. Transcriptional control by N⁴-methyl-cytosine may also exist¹³⁰.
- In multicellular eukaryotes, the DNA methylation pattern of the genome is reprogrammed during gametogenesis and during early embryonic development². In bacteria, reprogramming does not occur, and the DNA methylation pattern can be transmitted unaltered across generations. However, the acquisition and loss of DNA methyltransferase genes⁴¹ and recombinational shuffling of DNA methyltransferase domains^{77,33,143} can produce novel methylation patterns in bacterial genomes.
- In both bacteria and eukaryotes, DNA methylation controls the formation of phenotypic variants of genetically identical cells. However, DNA methylation-dependent formation of bacterial cell lineages can show programmed reversion (phase variation)^{15,27,93,111}.

only, and housekeeping functions were ascribed to solitary ('orphan') DNA methyltransferases not associated with R-M systems. This distinction is not valid anymore, as a number of R-M methyltransferases have been shown to have roles in addition to genome defence, including transcriptional regulation and the formation of phenotypic cell variants^{26–28}. The initial notion that solitary DNA methyltransferases were rare also did not hold up, as methylome analysis has shown that they are abundant²⁴. Finally, the semantics of the bacterial epigenome, which traditionally had centred on 6mA, has been extended to 5mC and 4mC.

In this Review, we summarize historic and recent developments in epigenetic signalling by DNA methylation, with an emphasis on its contribution to phenotypic heterogeneity in bacterial populations. This bias may be justified by the growing evidence that DNA methylation-mediated control of lineage formation is a widespread phenomenon in bacteria^{14,15,27,28}, whereas the control of physiological processes such as chromosome replication and mismatch repair may be restricted to certain bacterial clades⁸. An emphasis on phenotypic heterogeneity may be further warranted by theoretical analysis that has outlined its adaptive value^{29,30}. We also discuss, often in a speculative manner, evolutionary aspects of bacterial DNA methylation. Complementary information can be found in other reviews^{27,31–34}.

DNA methyltransferases

The formation of 6mA, 5mC and 4mC is catalysed by DNA methyltransferases, which transfer a methyl group from S-adenosyl-methionine (SAM) to DNA^{35,36}. Most

DNA methyltransferases are active as monomers³⁷. For the methylation reaction, the target base is rotated out ('flipped out') of the DNA helix and inserted into the catalytic pocket of the enzyme^{35,36}. Depending on the position of the target base in the double helix, DNA methyltransferases are classified into two groups: exocyclic (for 6mA and 4mC) and endocyclic (for 5mC)³⁷. Aside from this difference, all DNA methyltransferases show structural similarity, and the SAM-binding domain (or catalytic domain) is especially well conserved across kingdoms. By contrast, the target recognition domain (TRD) shows high variability in sequence and structure, which correlates with the manifold target specificities³⁸. Structural similarity and conservation of motifs between different groups of DNA methyltransferases may be indicative of a monophyletic origin, and diversification across kingdoms may have occurred upon the insertion and permutation of domains³⁹.

DNA methyltransferases associated with R-M systems are abundant, with an average of more than two R-M systems per genome^{40,41}. Species with higher numbers of R-M systems exist, and extreme examples are found in *Helicobacter pylori* strains that encode more than 50 R-M systems⁴². In addition, many bacterial and archaeal genomes harbour at least one orphan DNA methyltransferase²⁴. Orphan DNA methyltransferase genes are also found in about one fifth of phage genomes⁴³. Most orphan DNA methyltransferases are conserved at the genus level; the most highly represented are the Dam family in gammaproteobacteria and the CcrM family in alphaproteobacteria²⁴. Bioinformatic detection of a gene encoding a DNA methyltransferase does not imply that the enzyme is active, or even expressed: for example, transcription of the gene encoding the YhdJ methyltransferase of gammaproteobacteria is low or absent under laboratory conditions⁴⁴.

Most DNA methyltransferases recognize specific DNA targets: for instance, Dam and CcrM have a single target site (GATC and GANTC, respectively), whereas Dcm has two target sites, CCAGG and CCTGG⁵. Orphan DNA methyltransferases that have several TRDs (and thus are multispecific) have been described in *Bacillus subtilis* phages⁴³. Nonspecific DNA methyltransferases also exist, but they will not be discussed here⁴⁵.

The DNA methyltransferases of R-M systems methylate DNA in a distributive manner; that is, they methylate only one target upon DNA binding³⁷. Distributive DNA methylation can be seen as a crucial adaptation of these enzymes to their biological function, as the need to rebind DNA after each methylation event can slow down the methylation of phage DNA, thus facilitating restriction by the cognate R-M endonuclease³⁷. Orphan DNA methyltransferases are generally processive, and so methylate multiple targets without dissociating from DNA^{37,46}. However, the CcrM methyltransferase of alphaproteobacteria seems to be distributive⁴⁷. Processivity can vary depending on the genome context; for instance, certain DNA sequences that flank the GATC site can decrease the processivity of the Dam methyltransferase⁴⁸.

Bacterial DNA methyltransferases can be essential or dispensable^{7,13,49}. In the genus *Yersinia*, Dam-dependent

SOS regulon

Regulatory network that responds to DNA damage in gammaproteobacteria.

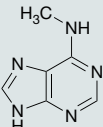
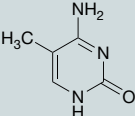
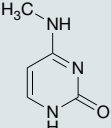
methylation is essential in *Yersinia enterocolitica*, and dispensable in two related species, *Yersinia pseudotuberculosis* and *Yersinia pestis*^{50–52}. Growth conditions can also determine whether a DNA methyltransferase is essential or dispensable: for instance, the CcrM methylase of *Caulobacter crescentus* is only essential in rich medium⁵³. The phenotypes of strains that lack or overproduce a DNA methyltransferase can be indicative of the physiological roles that DNA methylation might have in the species under study^{14,49,54,55}. Moreover, altered patterns of gene expression can provide evidence that the methylation state of promoters or regulatory regions controls the activity of the transcriptional machinery^{56,57}. However, altered gene expression does not necessarily imply direct transcriptional control, because indirect effects are possible. For instance, increased levels of SAM in DNA methylase mutants may induce changes in gene expression due to alterations in one-carbon metabolism⁵⁸. Similarly, transcriptome differences can reflect indirect control. For instance, upregulation of the SOS regulon in *dam* mutants of *E. coli* and *S. enterica* is not caused by Dam-dependent transcriptional control, but by DNA strand breakage^{59,60}. Enigmatic cases of post-transcriptional regulation by Dam-dependent methylation have also been described^{61,62}, and a tentative explanation is that Dam methylation may regulate transcription of genes involved in control of mRNA stability, mRNA translation or other post-transcriptional events.

Hemimethylation and demethylation

Bacterial DNA methyltransferases are active on both nonmethylated and hemimethylated DNA^{46,63}, thus acting as both writers of the methylome and copyists that transmit the methylation pattern of the genome to daughter cells. Because every round of DNA replication produces hemimethylated DNA from methylated DNA, methylome inheritance requires methylation of the newly synthesized DNA strand (FIG. 1). The duration of the hemimethylated state after DNA replication varies among bacterial clades, and a crucial factor is the availability of the DNA methyltransferase. In gammaproteobacteria, Dam methylase is synthesized at every stage of growth⁶⁴, and the hemimethylated state of GATC sites is short-lived because the Dam methylase trails the DNA replication fork at a short distance, estimated to be less than 10 kbp⁶⁵. By contrast, the genome of alphaproteobacteria remains hemimethylated during most of the cell cycle, because the CcrM methylase is only synthesized at the final stage of chromosome replication^{9,32,34}.

As a rule, hemimethylation after DNA replication is transient if the cognate DNA methyltransferase is available. However, at specific sites methylation of the daughter strand can be hindered by the binding of proteins that prevent DNA methyltransferase activity, causing methylation loss in a passive manner (FIG. 1). This phenomenon is well documented in *E. coli* and *S. Typhimurium*: stable hemimethylated GATC sites are formed when a DNA-binding protein protects hemimethylated DNA

Table 1 | Examples of writers and readers of the bacterial epigenome

Methylated base	Writer	Target sequence (5'–3')	Readers	Physiological process or phenotype under control
^N 6-methyl-adenine 	Dam	GATC	DnaA ¹⁰² , SeqA ⁸⁶ , MthLS ¹⁰⁵ , RNA polymerase ⁹¹ , transposase ^{91,134} , Lrp ^{69,88,89,163} , OxyR ^{70–72,94} , Fur ⁷³ and HdfR ^{74,75}	Chromosome replication and segregation ^{86,102,104} , nucleoid organization ^{87,164} , mismatch repair ¹⁰⁵ , transposition ^{91,134} , conjugal transfer ⁸⁸ , motility ⁵⁶ , synthesis and secretion of virulence determinants ^{11,12,56,62,165} , envelope structure ¹⁶⁶ , bacteriophage resistance ¹⁰⁷ and antibiotic resistance ¹⁵⁵
	CcrM	GATC	CcrA ⁹⁰ , MucR ^{76,167} and RNA polymerase ^{53,57}	Cell cycle control ^{32,34}
	ModA13	AGAAA	Unknown	Antibiotic resistance, epithelial cell invasion and biofilm formation ¹¹³
	SpnD39III	Allele-specific (six variants)	Unknown	Opacity (capsule synthesis) and virulence ¹²³
^C 5-methyl-cytosine 	Dcm	CCAGG and CCTGG	Unknown	General stress response ¹²⁸ and drug transport ¹²⁹
	JHP1050	GCGC	Unknown	Cell morphology, competence, outer membrane composition and copper resistance ¹²⁶
^N 4-methyl-cytosine 	M2.Hpy.All	TCTTC	Unknown	Virulence (adhesion to host cells) ¹³⁰

from Dam methylase activity^{6,14,66}. If Dam methylation hindrance persists during two consecutive rounds of genome replication, a nonmethylated GATC site is formed^{14,15}. Undermethylation requires that the processivity of Dam methylase is reduced, which occurs at GATC sites that are flanked by specific, A-T-rich DNA sequences^{48,67}. The effect of sequences outside the GATC site is not surprising, as DNA-bound Dam methyltransferase spans 10 bp⁶⁸. The transcription factors Lrp⁶⁹,

OxyR^{70–72}, Fur⁷³ and HdfR^{74,75} are examples of DNA-binding proteins that are involved in ‘passive’ erasure of methyl groups (see below). Undermethylated GATC sites have also been detected in alphaproteobacterial genomes, and an example of protection from CcrM methylation has been shown to involve a transcription factor, MucR⁷⁶. Passive demethylation may be a common phenomenon in the bacterial world: methylatable sites that lack methylation in one or both DNA strands are

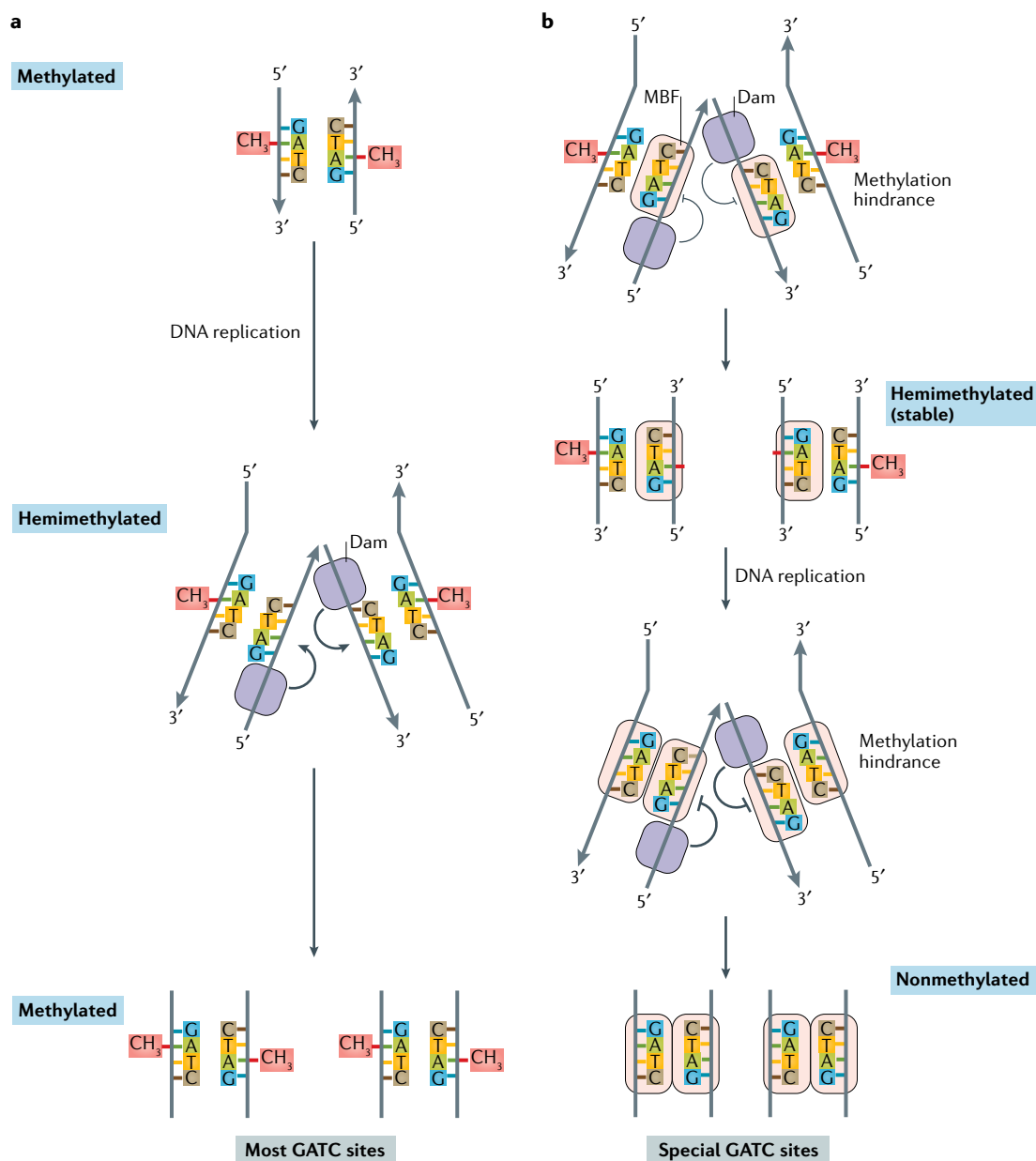


Fig. 1 | Methylation, hemimethylation and passive demethylation of GATC sites in gammaproteobacteria.

a | The bacterial genome is methylated at both DNA strands in the absence of DNA replication. Upon DNA replication, methylation is present in the template strand only (hemimethylation). Hemimethylation after DNA replication is transient as Dam, an orphan DNA methyltransferase that trails the DNA replication fork, targets GATC sites for methylation to restore the methylated state. This cycle occurs once per round of DNA replication and affects most GATC sites⁵. **b** | Binding of a protein that functions as a methylation-blocking factor (MBF) can prevent methylation of a hemimethylated GATC site by the Dam methyltransferase. If hindrance of methylation persists during two consecutive rounds of DNA replication, a nonmethylated GATC site is formed⁶⁶. DNA methylation hindrance by blocking factors occurs only at GATC sites flanked by DNA sequences that decrease the processivity of the Dam methylase⁴⁸ (not shown). Undermethylated (hemimethylated or nonmethylated) GATC site states can be transmitted to daughter cells in certain cases.

common in the genomes of gamma- and alphaproteobacteria^{24,77–80}, and some of those sites are located within protein-binding sites^{24,80}. In *E. coli*, the methylation state of certain GATC sites depends on growth conditions, which suggests that environmental effects can contribute to shape the methylome⁸¹.

‘Active’ DNA demethylation has been described in *E. coli*, and its physiological relevance remains to be fully understood. AlkB, a DNA repair protein that removes alkyl groups from nucleic acids, also targets 6mA and removes its methyl group⁸². Because AlkB expression is strongly induced upon DNA damage, widespread demethylation of 6mA may occur under such conditions. It is thus conceivable that the demethylation of physiologically relevant, critical GATC sites might contribute to survival, perhaps by delaying genome replication or cell division⁸².

Readout of methylation marks

The methyl group of 6mA, 5mC and 4mC protrudes from the major groove of the double helix, which provides a platform for DNA-binding proteins to bind to cognate nucleotide sequences. As a consequence, the methylation state of critical sites can either favour or hinder the interaction between a DNA-binding domain of the protein and a cognate DNA site⁷. In addition, base methylation can induce or enhance DNA curvature^{83–85}, which can further modulate DNA–protein interactions.

Inhibition of the activity of DNA restriction endonuclease by methylation of the target is a classic example of a methylation-sensitive DNA–protein interaction. However, discrimination of DNA methylation states can be far more complex. Certain methylation readers (for example, the GATC sequestration factor SeqA) can bind both methylated and hemimethylated DNA targets, albeit with different affinities^{86,87}. Other proteins (for example, the transcription factor Lrp) bind both hemimethylated and nonmethylated targets, but the binding patterns and the affinities are different^{88,89}. Another remarkable bias in protein binding to hemimethylated sites is strand specificity, which can favour binding to only one of the hemimethylated DNA species produced by DNA replication^{89–91}. An unproven but reasonable speculation is that the overall methylation state of the genome may also influence the binding of proteins to specific regions, by moulding nucleoid topology^{34,92}.

Especially complex DNA–protein interactions occur at regions that harbour several protein-binding sites, which in turn contain methylatable motifs^{6,15,66}. In such cases the protein-binding pattern determines the methylation state, because methylation hindrance causes passive demethylation of specific sites. In turn, the methylation pattern determines the protein-binding pattern. The same protein is thus the eraser of methyl groups and the reader of the resulting DNA methylation pattern. In the imaginary example presented in FIG. 2a, the existence of two methylatable binding sites permits alternative, mutually exclusive binding patterns, and each binding pattern determines a distinct DNA methylation pattern, because methylation is hindered at the sites bound by the regulator^{6,66}. Such combinations of methylated and

nonmethylated sites can be heritable, thus transmitting epigenetic states to daughter cells^{6,15}. In gammaproteobacteria, Dam-dependent methylation patterns are found at the transcriptional control regions of bistable loci, and the elements involved in the formation of OFF and ON states are more complex than those in FIG. 2a. For instance, the *E. coli pap* operon contains six binding sites for Lrp and two GATC sites⁹³; the *S. enterica opvAB* and *gtr* operons contain four OxyR binding sites and four GATC sites^{72,94}; the *E. coli sciI* locus contains three Fur binding sites and three GATCs⁷³; and the *S. Typhimurium std* operon contains one binding site for HdfR and three GATC sites⁷⁵ (FIG. 2b). In all cases, the ON and OFF states are dictated by the binding pattern of the cognate protein. Binding causes passive demethylation, which can increase the affinity of the regulator towards its cognate site (or sites), thus generating a positive feedback loop that stabilizes the epigenetic state, which can sometimes be heritable.

Certain bistable loci that are controlled by Dam-dependent methylation switch between alternative states, and the system is thus reversible (‘phase-variable’; see below)^{15,66}. Switching between states requires DNA replication with concomitant formation of hemimethylated DNA, and a crucial factor that determines the frequency of switching is the affinity of the DNA-binding protein for its binding sites^{72,95} (FIG. 2a). Ancillary factors may affect switching: for instance, in the *pap* operon a protein named PapI is necessary for switching from OFF to ON and for maintenance of the ON state⁹⁵.

Frequencies of ON–OFF switching are locus specific. For instance, under laboratory conditions the Gtr^{OFF} and Gtr^{ON} subpopulations of *S. enterica* have roughly identical sizes⁹⁶, whereas the OpvAB and Std switches are lopsided towards the OFF state^{72,97}. However, switching can be under environmental influence: for instance, the Std^{OFF} and Std^{ON} subpopulations have different sizes in the laboratory and in the animal intestine^{97,98} (see below). Additional evidence for environmental control has come from the observation that housekeeping proteins responsive to environmental cues can alter switching frequencies. Examples include the global regulator CRP⁹⁹, the signal transduction proteins CpxAR¹⁰⁰ and the nucleoid proteins H-NS and HU^{72,101}.

Cell cycle signalling by DNA methylation

In gammaproteobacteria, the initiation of chromosome replication and segregation of newly replicated nucleoids is controlled by readers that discriminate methylation from hemimethylation. One such reader is DnaA, which initiates chromosome replication if GATC sites within the origin of replication (*oriC*) are methylated¹⁰². Another reader of the methylation state is SeqA, which binds hemimethylated GATC sites at the *oriC* and delays Dam methyltransferase activity⁸⁶. Transient hemimethylation of the *oriC* by SeqA also represses transcription of the nearby *dnaA* gene, which contains GATC sites in the promoter region¹⁰³. In addition, the binding of SeqA to hemimethylated GATC sites governs the condensation and segregation of newly replicated nucleoids¹⁰⁴. Transient hemimethylation upon replication fork passage is also a signal for repair of

mismatched nucleotide pairs, which relies on the ability of the MutHLS complex to cleave the daughter strand at hemimethylated GATCs¹⁰⁵.

In *C. crescentus*, the initiation of chromosome replication by DnaA requires GATC methylation at the replication origin (*Cori*)^{32,34}. However, *Cori* remains spontaneously hemimethylated, without sequestration, because the CcrM methylase becomes available only at the end of every round of chromosome replication³². Hence, the DnaA proteins of gamma- and alphaproteobacteria read analogous signals at *oriC* and *Cori*, respectively, but the mechanisms that control methylation of the replication origin are different^{7,32}.

Reading of DNA hemimethylation signals by RNA polymerase and transcription factors can activate

transcription upon passage of the replication fork. Classic examples in gammaproteobacteria are the transposase (*tnp*) gene of insertion element IS10 (REF.⁹¹) and the plasmid-borne *traJ* gene, which encodes an activator of conjugal transfer^{88,89}. Because the hemimethylation state is transient, the expression of *tnp* and *traJ* is short-lived, and synthesis of the corresponding gene products is restrained. An additional sophistication of *tnp* and *traJ* hemimethylation readout is asymmetry: the activation of *tnp* and *traJ* transcription takes place in only one of the hemimethylated DNA species produced by DNA replication^{89,91}. The adaptive value of strand-specific transcription of *tnp* and *traJ* may be tentatively understood: asymmetric reading restrains synthesis of dangerous (transposase) or energetically

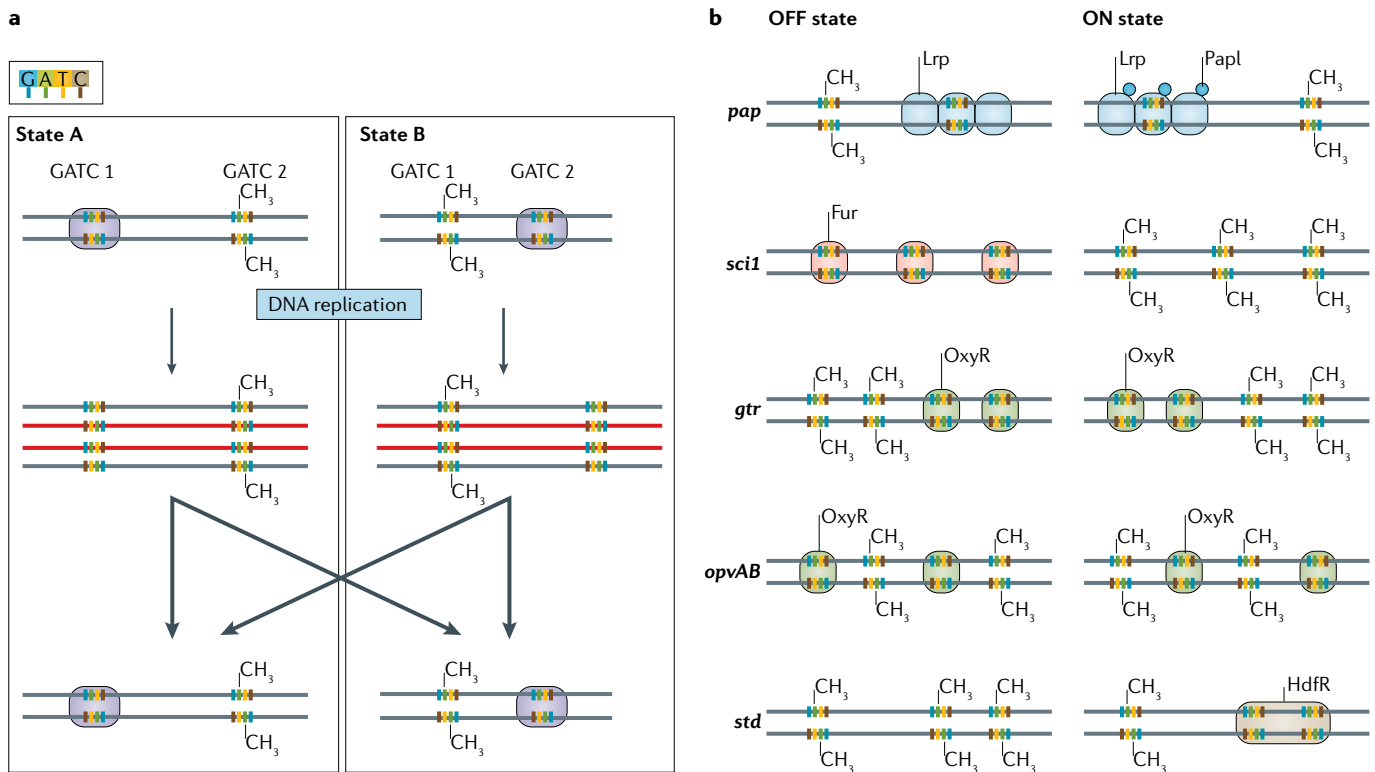


Fig. 2 | Control of bistable loci by the formation of alternative Dam methylation patterns. **a** | Shown is an imaginary locus that harbours two protein-binding sites, each containing a GATC (GATC 1 and GATC 2). Two binding patterns are possible, and each binding pattern causes passive demethylation of a specific GATC site because methylation is hindered at the sites bound by the regulator. Nonmethylation can increase the affinity of the protein for its cognate binding site, thus creating a positive feedback loop that stabilizes the DNA methylation pattern and makes it heritable. However, the system can be reset when hemimethylated DNA substrates are formed upon DNA replication. Switching occurs if the protein moves from one cognate site to the other, and a major factor that determines the frequency of switching in each direction is the affinity of the protein towards each binding site^{72,95}. Ancillary factors (not shown) can skew switching, either in a stochastic manner or in response to environmental signals^{100,101}. **b** | Epigenetic states of bistable loci of *Escherichia coli* (*pap* and *sci1*) and *Salmonella enterica* (*gtr*, *opvAB* and *std*) under the control of Dam methylation. The diagrams show regulatory regions upstream of the promoters, with the promoter-proximal side on the right. Transcription is repressed in the OFF state and activated in the ON state. In all cases, the regulatory region contains binding sites for a transcription factor:

six binding sites for Lrp in *pap*⁹³, four for OxyR in *gtr*⁹⁴ and *opvAB*¹⁰⁶, three for Fur in *sci1* (REF.⁷³), and one binding site for HdfR in *std*⁷⁵. The transcription factor binding sites contain GATC sites: two in *pap*⁹³, three in *sci1* (REF.⁷³) and *std*⁷⁵, and four in *opvAB*¹⁰⁷ and *gtr*⁹⁴. The methylation state of such sites is dictated by the binding pattern of the transcription factor: GATCs located within sites bound by the transcription factor undergo passive demethylation, while GATCs within unbound sites are methylated. This mutual exclusion between Dam methylation and transcription factor binding produces distinct patterns of DNA methylation in the regulatory region of OFF and ON cells. In *Pap*^{OFF} cells, the upstream GATC site is methylated and the downstream GATC is nonmethylated, and the opposite GATC methylation pattern is found in the *Pap*^{ON} lineage⁹³. In the regulatory regions of *gtr* and *opvAB*, cells in OFF and ON states display opposite Dam methylation patterns involving two methylated and two nonmethylated GATC sites^{72,94}. The formation of nonmethylated GATC sites is likewise observed in *Sci1*^{OFF} cells as the consequence of Fur binding⁷³, and in *Std*^{ON} cells as the consequence of HdfR binding⁷⁵. Under laboratory conditions, the formation of ON and OFF subpopulations is reversible (phase variable) in *pap*¹⁰, *gtr*⁹⁴, *opvAB*¹⁰⁶ and *std*⁹⁷, but not in *sci1* (REF.⁷³). In *pap*, switching to ON also requires an ancillary factor, PapI⁹⁵.

expensive (conjugation) cell products. Asymmetric transcription from a hemimethylated promoter has also been observed in the *creS* gene of *C. crescentus*⁷⁶, which suggests that the transcription factor involved (GrcA) may be a hemimethylation reader with DNA strand specificity. An attractive speculation is that asymmetric reading of GATC hemimethylation may contribute to differentiation of the two cell types³⁴.

Bistability and phase variation

Heritable DNA adenine methylation patterns. The formation of alternative Dam-dependent methylation patterns, a phenomenon discussed above, controls the expression of bistable loci and produces phenotypic variants of genetically identical cells in gammaproteobacteria⁶⁶. Dam-dependent bistability can be phase-variable¹⁵. Paradigms of Dam-dependent phase variation in *E. coli* are the *pap* operon of uropathogenic strains, which encodes fimbriae for adhesion to the urinary epithelium⁹³, and the *agn43* gene, which encodes a non-fimbrial adhesin involved in bacterial auto-aggregation^{70,71}. Descriptions of Dam-dependent control of *pap* and *agn43* phase variation can be found in other reviews^{6,7,66,93} and will not be discussed here.

Natural isolates of non-typhoidal *S. enterica* carry one to four operons that encode glucosyltransferases, which add sugars to the lipopolysaccharide O-antigen⁹⁶. Such operons, named *gtr*, are associated with prophages or prophage remnants present in the genome, and each operon may produce a distinct modification of the O-antigen. The resulting variants of the bacterial surface may adapt *Salmonella* to interact with different host receptors, and they may also alter phage susceptibility⁹⁶. The expression of *gtr* is phase-variable, and the *Gtr*^{OFF} and *Gtr*^{ON} subpopulations harbour alternative patterns of methylated and nonmethylated GATC sites within a regulatory region upstream of the *gtr* promoter⁹⁴. The transcription factor OxyR is both an eraser of methyl groups and a reader of the resulting *gtr* methylation patterns: OxyR binding to upstream cognate sites prevents methylation of GATC₁ and GATC₂ and activates transcription, whereas binding to downstream sites prevents methylation of GATC₃ and GATC₄ and represses transcription^{94,96} (FIG. 2b).

Another phase-variable operon under the control of Dam-dependent methylation and OxyR is *opvAB*, whose products are cytoplasmic membrane proteins that reduce the length of the *S. enterica* lipopolysaccharide O-antigen¹⁰⁶. As in *gtr*, the control of phase variation is transcriptional. A regulatory region upstream of the *opvAB* promoter contains four OxyR binding sites and four methylatable GATC motifs; *OpvAB*^{OFF} and *OpvAB*^{ON} cell lineages display opposite patterns of OxyR binding, which in turn generate opposite patterns of GATC methylation⁷² (FIG. 2b). Transcriptional switching is skewed towards the *OpvAB*^{OFF} state, and the *OpvAB*^{ON} lineage comprises only 0.3% of cells¹⁰⁶. Shortening of the O-antigen renders *OpvAB*^{ON} cells resistant to phages that use the O-antigen as a receptor. *OpvAB*^{ON} cells are unable to infect animal hosts¹⁰⁷; however, as soon as phage challenge ceases, phase variation produces *OpvAB*^{OFF} cells, and lopsided switching towards the OFF

state quickly reproduces a population in which >99% of cells are *OpvAB*^{OFF} (FIG. 3a). The phase variation of the *opvAB* operon may thus be viewed as a trade-off between virulence and bacteriophage resistance¹⁰⁷.

Another phase-variable locus, *std*, provides a unique example of the power of Dam-dependent phase variation to produce phenotypic diversity. The *std* operon is found in the genome of *S. Typhimurium* and encodes fimbriae that facilitate adhesion to the mucus layer of the large intestine^{98,108}. In addition, the *std* operon encodes two transcription factors, StdE and StdF, that are activators or repressors of more than one hundred genes⁹⁷. Pleiotropic control thus splits the bacterial population into two subpopulations that differ in multiple phenotypic traits, a phenomenon that may be viewed as a rudimentary differentiation program⁹⁷ (FIG. 3b). The molecular mechanisms that control *std* phase variation are more complex than those in other Dam methylation-dependent loci: *std* transcription is activated by HdfR, a transcription factor of the LysR family, and the transcription of *hdfR* is activated by an *std* product, StdE⁷⁵. The binding of HdfR upstream of the *std* promoter is hindered by Dam-dependent methylation, and HdfR binding prevents methylation of two GATC sites in the regulatory region⁷⁵ (FIG. 2b). In addition, the formation of Std^{ON} cells requires autogenous activation of transcription by StdF, which binds the *std* promoter in a Dam methylation-independent fashion⁷⁵.

Phase-variable DNA methyltransferases of R-M systems.

Two decades ago, a *Haemophilus influenzae* gene (*mod*) homologous to genes encoding the DNA adenine methyltransferases of type III R-M systems was found to undergo phase variation due to contraction and/or expansion of nucleotide repeats¹⁰⁹. Another phase-variable *mod* gene belonging to a type III R-M system was thereafter described in *H. pylori*, and phase variation was found to be caused by length variation of a homopolymeric nucleotide tract in the 5'-coding region of a gene located upstream of *mod*¹¹⁰. The type III R-M Mod enzyme of *H. influenzae* turned out to be more than just a modification enzyme involved in genome protection: transcriptome analysis identified genes that were either up- or downregulated in a *mod* mutant, which indicates that adenine methylation of specific genome sequences controlled the expression of specific genes. Switching in the availability of active DNA methyltransferase thus produced a phase-variable regulon ('phase-varion')¹¹¹. An additional, remarkable observation was that the restriction enzyme of certain type III R-M systems was inactive in a number of *Haemophilus* strains, which suggests that such R-M systems had evolved into epigenetic regulators of gene expression¹¹². Further studies described 6mA phasevarions in *Neisseria meningitidis* and *Neisseria gonorrhoeae*¹¹³. Phase variation involved variations in the length of simple sequence repeats (FIG. 4), and the loci under the control of Mod-mediated methylation included genes with roles in envelope structure, virulence and stress responses¹¹⁴. The power of phasevarions as sources of epigenetic variation is remarkable: in contrast with the phase variation systems that generate heterogeneity of a single phenotypic trait,

Homopolymeric nucleotide tract
DNA region that contains only
AT or GC nucleotide pairs.

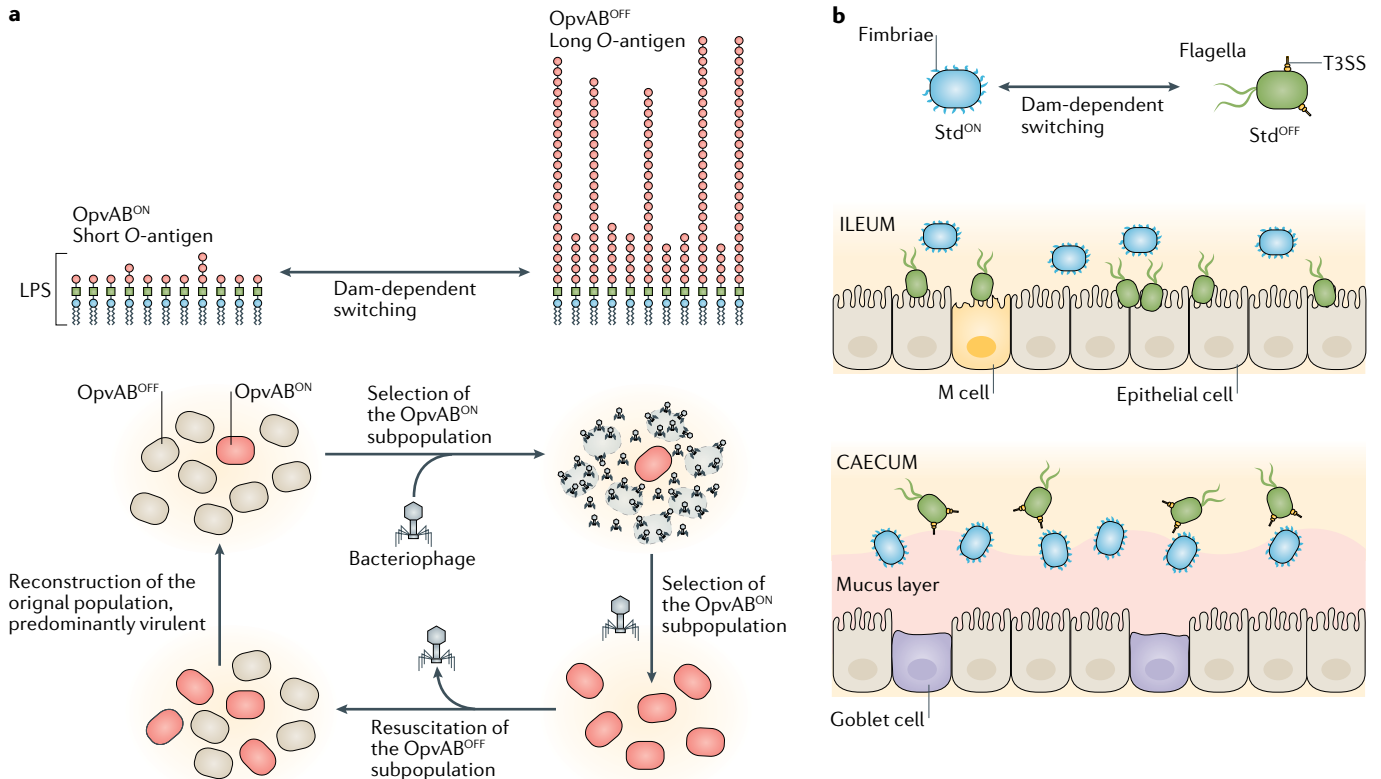


Fig. 3 | Formation of subpopulations controlled by Dam-dependent methylation. a | Phase-variable transcription of the *Salmonella enterica* *opvAB* operon produces a lineage of OpvAB^{OFF} cells with long O-antigen chains in the lipopolysaccharide (LPS) (OpvAB^{OFF}) and a lineage with shorter O-antigen chains (OpvAB^{ON})¹⁰⁶. Shortening of the O-antigen renders the OpvAB^{ON} lineage avirulent but resistant to bacteriophages (many *Salmonella*-targeting phages use the O-antigen as a receptor). In the presence of a phage, the OpvAB^{OFF} subpopulation will be eliminated, while the OpvAB^{ON} subpopulation will survive. When the phage challenge ceases, OpvAB^{OFF} cells produced by phase variation will survive, and virulence will be regained. However, a small subpopulation of avirulent OpvAB^{ON} cells will sustain preadaptation to future phage challenge¹⁰⁷. **b** | Phase-variable transcription of the *std* operon produces subpopulations of Std^{OFF} and Std^{ON} cells⁹⁷. The Std^{OFF} lineage is motile (producing flagella) and synthesizes the type 3 secretion system (T3SS), promoting invasion of epithelial cells in the ileum of infected animals, which is the initial stage of systemic *Salmonella* spp. infection^{97,162}. In turn, the Std^{ON} lineage produces fimbriae that permit adhesion to the mucus layer of the caecum in the large intestine^{98,108} but harbours neither flagella nor a T3SS⁹⁷. Adhesion of the Std fimbriae to the caecal mucosa increases inflammation⁹⁸, which promotes the survival of *Salmonella*. Std^{OFF} cells can thus cause acute infection, whereas Std^{ON} cells cause chronic infection, and *std* phase variation can be viewed either as a division of labour or as a bet-hedging strategy.

bacterial lineages under phasevarion control can differ in multiple phenotypic traits.

Additional examples of phase-variable type III R-M systems have been described in the past decade^{33,115–117}, and bioinformatic analysis predicts that nearly one fifth of type III *mod* genes may be phase variable, as judged from the presence of sequence repeats¹¹⁷. In certain bacterial species (for example, *Neisseria* spp.), *mod* alleles show high homology except in the target recognition domain, thus providing evidence for rapid evolution by recombinational shuffling¹¹³. Target diversification, together with frequent horizontal transfer of *mod* genes¹¹⁸, may further promote phenotypic diversity¹¹⁹.

A DNA adenine methyltransferase of a phase-variable type II R-M system has been shown to regulate gene expression and virulence in *Campylobacter jejuni*, where phase variation is controlled by the insertion or deletion of repeat units. However, it remains to be established whether or not gene expression control involves

the presence of 6mA at promoters or transcription control regions¹²⁰.

Phase variation of DNA adenine methyltransferases belonging to type I R-M systems was first identified in *Mycoplasma pulmonis*¹²¹, and later in other human pathogens^{27,122}. A frequent mechanism of switching is recombination between inverted repeats of type I R-M (*hsd*) genes, which can be present in multiple copies, complete or incomplete, in a genome²⁷ (FIG. 4). DNA inversion by site-specific recombination and the expansion and/or contraction of single sequence repeats can also mediate the switching of *hsd* genes^{27,33} (FIG. 4). An example of the contribution of type I R-M systems to phase variation has been described in *Streptococcus pneumoniae*, in which genetic rearrangements generate six 6mA variants of a type I DNA methyltransferase that controls the transparent–opaque transition^{123–125}. Each DNA methyltransferase variant generates a distinct methylation pattern in the genome, which results in different gene

Transparent–opaque transition

Formation of phenotypic variants (phase variation) of *Streptococcus pneumoniae* that is involved in pneumococcal carriage and invasive infection.

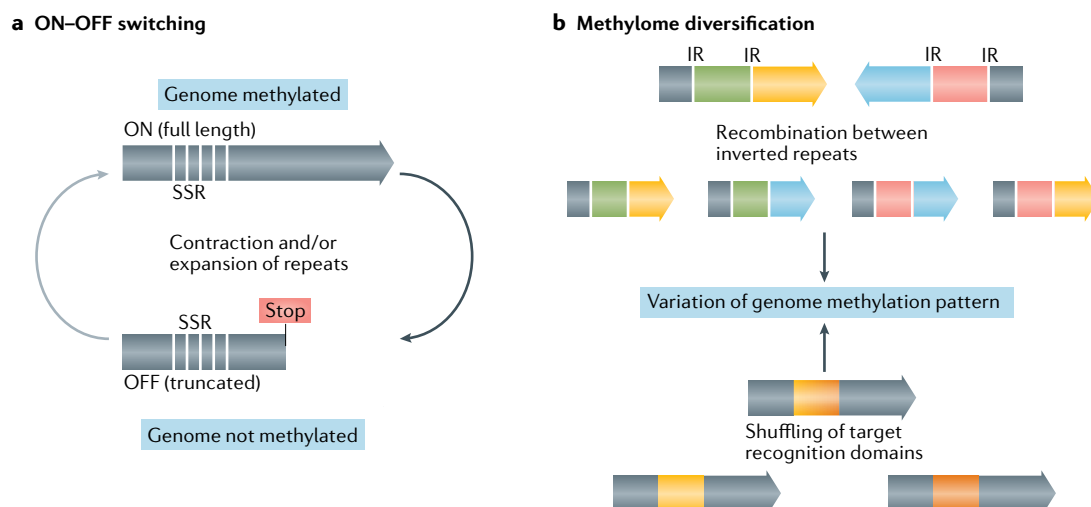


Fig. 4 | **Mechanisms of switching and methylome diversification in phase variations.** **a** | Phase variation (ON–OFF switching) by the expansion and/or contraction of single sequence repeats (SSRs). Frameshifting in the coding sequence of a DNA methyltransferase gene causes premature termination, as a stop codon is introduced, generating a cell lineage whose genome lacks DNA methylation^{27,33,114}. However, further frameshifting events can either revert or suppress the frameshift, and the restoration of a functional reading frame can permit DNA methyltransferase synthesis and genome methylation. If base methylation affects the DNA–protein interactions that are crucial for gene expression, each bacterial lineage will show a distinct expression pattern of methylation-sensitive loci¹¹⁴. **b** | Formation of DNA methyltransferase variants by recombination between inverted repeats (IRs) or between target recognition domains. DNA methyltransferase variants with distinct specificities generate allele-specific patterns of genome methylation, which can in turn have phenotypic consequences²⁷.

expression profiles and produces pneumococcal lineages with different virulence capacities^{123–125}.

Transcriptional control by 5mC and 4mC

JHP1050 is a 5mC DNA methyltransferase derived from an ancestral R-M system that is highly conserved among *H. pylori* strains¹²⁶. Mutants that lack JHP1050 show alterations in adherence to host cells, natural competence for DNA uptake, bacterial cell shape and susceptibility to copper¹²⁶. These phenotypes correlate with altered patterns of gene expression, and site-directed mutagenesis of GCGC targets at promoter regions has provided evidence for direct transcriptional control by 5mC¹²⁶. The formation of 5mC may further influence gene expression in an indirect manner, perhaps by changes in DNA topology, as was previously proposed for 6mA⁹².

In *Vibrio cholerae*, the VhcM orphan 5mC methyltransferase is necessary for optimal growth, both in vitro and during infection, and modulates stress responses¹²⁷. Transcriptome changes in *vhcM* deletion mutants may be a consequence of direct transcriptional control, but indirect effects are also conceivable¹²⁷. A similar situation may apply to upregulation of the general stress response in *E. coli* mutants lacking the Dcm 5mC methyltransferase¹²⁸, as well as to RpoS-dependent control of a drug transporter¹²⁹.

The literature also contains one example of epigenetic signalling by 4mC: M2.Hpy.AII, a DNA methyltransferase that introduces a methyl group at the N⁴ position in the first C of TCTTC motifs, controls gene expression and virulence-related traits in *H. pylori*¹³⁰. As in other studies discussed above, transcriptional control may be either direct or indirect¹³⁰.

DNA methylation and bacterial evolution

DNA methylation is both a conservative and an innovative force in bacterial evolution. R-M systems function as barriers for the acquisition of genetic elements by horizontal transfer^{26,40}. Furthermore, R-M systems can limit recombination between bacterial clades that harbour distinct R-M repertoires^{131,132}. A likely mechanism is cleavage of incoming DNA molecules, with a concomitant shortening of recombination tracts. These conservative roles are strengthened by the behaviour of R-M systems as toxin–antitoxin addiction modules: if an R-M system is lost, the decrease in the number of DNA methyltransferase molecules may permit restriction, and a single chromosome break can be sufficient to cause cell death^{40,133}. Among orphan methyltransferases, Dam contributes to genome maintenance by the repression of transposable elements^{91,134} and prophages^{135,136}, as well as by reduction of the mutation rate via mismatch repair¹⁰⁵. Interestingly, Dam methyltransferase can also be viewed as an additive component of the genome, as its loss causes genome instability upon DNA strand breakage^{59,137}.

A major role of both Dam and phase-variable DNA methyltransferases is the formation of phenotypic lineages, which enables division of labour in a community or prepares the community for future changes in the environment (bet-hedging)¹³⁸. An organism can only have a limited set of traits, and improvement of a given phenotype comes at the expense of another phenotype¹³⁹. Lineage formation solves this problem by allotting specific phenotypes to each cell type in a manner that can benefit the whole population. Formation of epigenetic cell variants may also provide a strategy for mutation

avoidance: mutations are often deleterious, and their accumulation in a population can decrease fitness^{140,141}. In fact, bacterial populations with high mutation rates adapt quickly to a given environment but are outcompeted by non-mutators in the long run¹⁴². By contrast, non-mutational variation can adapt bacterial populations to environmental challenge with lower fitness

costs. An example is provided by the Dam methylation-dependent OvpAB bet-hedging system of *S. enterica*, which confers bacteriophage resistance by modification of the O-antigen¹⁰⁷ (see above). In an *opvAB* deletion mutant strain, bacteriophage-resistant mutants can be isolated; however, such mutants often harbour mutations that impair fitness¹⁰⁷.

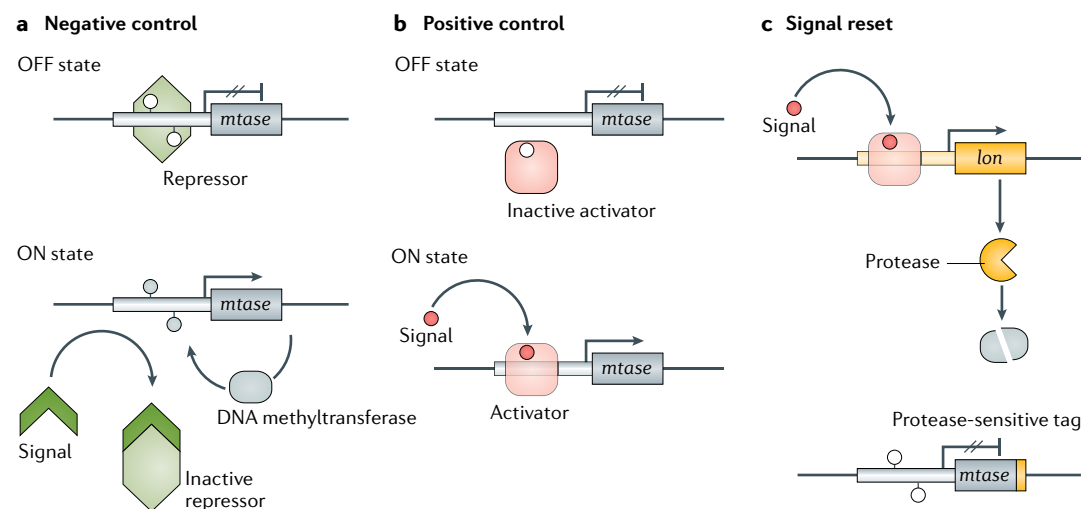
Box 2 | Applications of DNA methylation in biotechnology and synthetic biology

Lack of DNA methylation attenuates virulence, sometimes severely, in a large number of bacterial pathogens¹³. In species in which DNA methylation is essential and mutants are not viable, the attenuation of virulence has been observed following the overproduction of DNA methyltransferases¹³. This relationship between DNA methylation and virulence has fostered novel preventive and therapeutic approaches to combat bacterial infections. For instance, live vaccines against *Salmonella* spp. have been developed using strains that carry *dam* mutations, alone or combined with other mutations that attenuate virulence^{159,160}. Another example is an oral *Yersinia pseudotuberculosis* vaccine, which has been shown to be protective in mice⁵². A different approach involves the use of DNA methyltransferase inhibitors as antibacterial agents¹⁵². In gammabacterial pathogens in which Dam-dependent methylation is dispensable, the inhibition of Dam methylase activity can produce phenocopies of *dam* mutants, thus reducing virulence without affecting viability. As a consequence, the selection of inhibitor-resistant variants may not be strong. An additional advantage is that inhibitors of DNA adenine methylation may be efficient against various bacterial pathogens, as DNA adenine methyltransferases show structural similarity in different bacterial clades³⁸. Inhibitors of the DNA adenine methyltransferases of alphaproteobacteria and of Gram-positive bacteria have been also described^{153,154}. The inhibition of DNA adenine methylation was initially considered harmless for the mammalian host. However, the recent discovery of DNA adenine methylation in the human genome¹⁶¹ raises the concern of possible side effects.

A recent study has suggested the possibility of using Dam methylase inhibitors to enhance the therapeutic activity of antibiotics¹⁵⁵. Lack of Dam methylation increases the bactericidal activity of β -lactams, because exposure to the antibiotic causes oxidative damage, which in turn induces SOS-dependent error-prone repair. In wild-type *Escherichia coli*, Dam-dependent mismatch repair corrects the errors. However, in the absence of Dam methylation, the MutHLS mismatch repair system generates double-strand breaks in the DNA. Such breaks further activate the SOS response, thus generating a toxic feedback loop that potentiates the lethal action of the antibiotic¹⁵⁵.

DNA methylation may also have application in phage therapy: the use of phages that encode DNA methyltransferases can be expected to boost the elimination of infectious bacteria, by delaying the ability of the host R-M systems to kill the incoming phage⁴³.

An appealing but largely unexplored application of bacterial DNA methylation is the development of DNA methylation-based biosensors (see the figure). An example is provided by a recent study that describes synthetic memory systems that are able to store information in *E. coli* in the form of CcrM-dependent methylation patterns¹⁵⁶. In the negative control system (see the figure, part a), signal detection dislodges a repressor from the control region of a DNA methylase gene (*mtase*), and methylation of the control region prevents repressor binding. The resulting feedback loop will keep the sensor active even if the signal disappears, and the DNA methylation pattern will be transmitted to daughter cells, thus providing memory of a past event¹⁵⁶. In the positive control system (see the figure, part b), signal detection enables a transcription factor to activate the expression of a DNA methylase gene. In both systems, genome methylation will occur in response to the signal. The addition of a fluorescent protein gene downstream of the DNA methylase gene (not shown in the figure) can help detect sensor activation. Variants of these systems have been engineered to detect various environmental stimuli including heat, DNA-damaging agents and nutrients. A device for signal reset can be added to memory sensors (see the figure, part c): induced synthesis of a protease (Lon) permits the degradation of an engineered DNA methylase variant that contains a protease-sensitive tag¹⁵⁶.



Although Dam methylation-dependent switches and phasevarions may have analogous roles in the formation of bacterial lineages, they must be viewed as the products of distinct evolutionary strategies, each with advantages and drawbacks. Dam-dependent switches are the products of long-term co-evolution with the core genome, and the formation of Dam-dependent methylation patterns may be an especially robust mechanism for transmission of epigenetic memory to daughter cells. Furthermore, crosstalk with the housekeeping machinery may permit adjustment of the switching frequency in response to environmental cues^{72,100,101}. However, integration into host regulatory networks may heavily constrain further evolution. By contrast, stochastic switching of phasevarions may be prone to perturbation but at the same time be more evolvable, and independence from host genome control may further increase evolvability. Recombination between DNA methyltransferase alleles may be a powerful mechanism for the generation of DNA methyltransferase alleles with novel TRDs, especially in species that harbour multiple DNA methyltransferase genes^{113,118}.

High rates of horizontal transfer of DNA methylase genes, either R-M or orphan, can further promote bacterial diversification^{40,41}. Indeed, DNA-recognition domains seem to be highly mobile¹⁴³. In R-M systems, a change of DNA methylation specificity can permit the acquisition of novel foreign elements, including prophages, transposons, plasmids and other mobile elements, perhaps accompanied by the loss of resident, old components of the peripheral genome⁴¹. Furthermore, the acquisition of a novel methylation pattern may modify the transcriptome if the transcriptional machinery of the recipient cell contains appropriate readers. The fate of the new-born pattern of epigenetic regulation will be determined by natural selection, and it might instantly produce a novel bacterial clade, in the case of high adaptive value. Shuffling and exchange of TRD modules may thus be viewed as a trial-and-error game for bacterial diversification.

The use of 5mC as an epigenetic mark has an enigmatic side because 5mC is unstable in all cells, including bacteria, and 5mC deamination produces thymine¹⁴⁴. In the resulting T:G mismatches, discrimination of the incorrect base is not feasible, as both T and G are normal bases. If left unrepaired, a GC to AT transition will occur upon DNA replication, and one of the daughter DNA molecules will contain a base pair substitution. Gammaproteobacteria possess a repair system (very short patch repair) that corrects T:G mismatches by replacing T with C¹⁴⁵. However, a system of this kind is absent in many bacterial clades: for instance, 5mC deamination contributes to the high mutation frequency of *H. pylori*¹²⁶. One can speculate that a high mutation rate may have adaptive value in certain species and

may represent a burden in others. However, asymmetric mutation towards AT seems to be a universal phenomenon in bacterial genomes¹⁴⁶, and the bias may be explained by a high rate of 5mC deamination¹⁴⁷.

Conclusions and future perspectives

Conservation of the DNA methyltransferase structure suggests its early evolutionary origin³⁸, and the widespread methylation of bacterial and archaeal genomes may be indicative of positive selection during evolution²⁴. A conceivable scenario is that DNA methylation arose as a genome defence mechanism associated with restriction⁴⁰. An alternative possibility is that restriction enzymes did not initially evolve to destroy foreign DNA molecules, but to guarantee that DNA methyltransferases remained active²⁴. High rates of horizontal transfer of DNA methyltransferase genes have contributed to the spread of DNA methylation in the bacterial world⁴¹, and exchange and shuffling of DNA methyltransferase genes have promoted methylome diversification^{24,27,118,143}.

During evolution, and perhaps very early, the addition of epigenetic information to the nucleotide sequence turned out to confer adaptive traits to bacterial cells. One such trait is the ability to form phenotypic cell variants, often in a reversible manner^{15,27,111}. An asset of reversibility is the ability to restore phenotypic heterogeneity after a bottleneck, a circumstance that is especially frequent inside animal hosts²⁷. The abundance of phasevarions in bacterial pathogens may support this view^{27,33,114}.

In certain bacterial clades, long-term co-evolution with the host genome has permitted DNA methyltransferases to participate in the control of housekeeping processes, including the cell cycle^{32,34}, chromosome replication and segregation^{6–8}, DNA repair¹⁰⁵ and the secretion of virulence determinants¹³. Functional analysis of bacterial methylomes may reveal novel roles in years to come.

Modification of the host epigenome by bacterial pathogens is a well-established phenomenon, the significance of which remains to be understood^{148–150}. Epigenetic memory of a bacterial infection might benefit either the host (for example, by boosting immune responses), the pathogen (for example, by weakening immunity) or both organisms (for example, by facilitating commensal or symbiotic co-existence). An intriguing possibility is that DNA methyltransferases of intracellular pathogens may directly methylate the eukaryotic genome¹⁵¹.

Future applications of bacterial DNA methylation in biotechnology may include the use of inhibitors of DNA adenine methylation as antibacterial drugs^{152–155} and the design of synthetic switches, sensors and memory devices^{156,157} (BOX 2).

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Author contributions

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